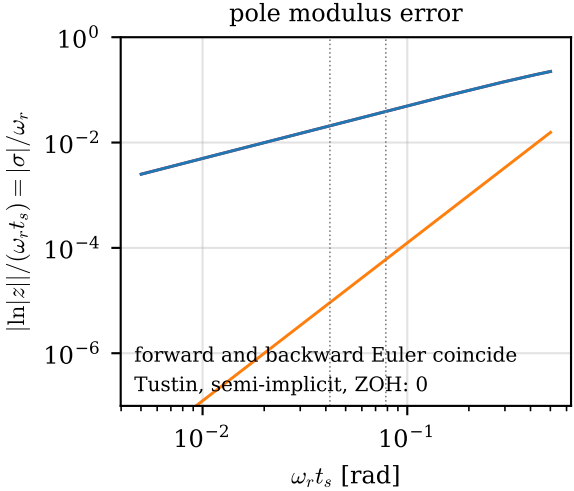
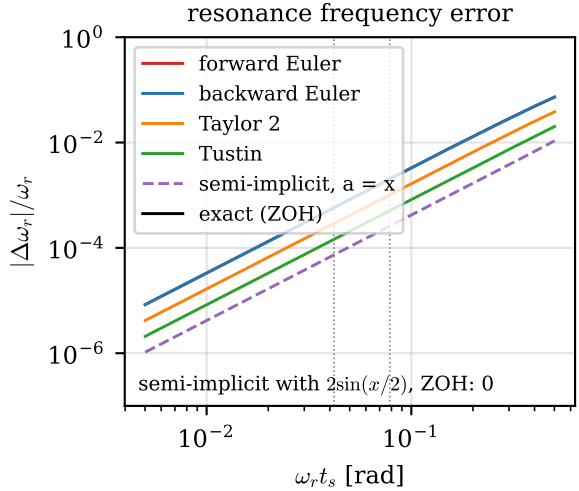

pole modulus error
10^0
10^{-2}
10^{-4}
10^{-6}
$|\ln|z||/(\omega_r t_s) = |\sigma|/\omega_r$
forward and backward Euler coincide
Tustin, semi-implicit, ZOH: 0
10^{-2}
10^{-1}
$\omega_r t_s$ [rad]


resonance frequency error
10^0
10^{-2}
10^{-4}
10^{-6}
$|\Delta\omega_r|/\omega_r$
forward Euler
backward Euler
Taylor 2
Tustin
semi-implicit, a = x
exact (ZOH)
semi-implicit with $2\sin(x/2)$, ZOH: 0
10^{-2}
10^{-1}
$\omega_r t_s$ [rad]